

Environment and development discourse, Environment issues

Environment and development discourse

Environment:

- Environment refers to the surroundings and conditions in which humans, animals, and plants live and operate.
- It includes both **living (biotic)** and **non-living (abiotic)** components.
- Major components include air, water, land, plants, animals, microorganisms, and their interactions.
- Human survival and economic activities depend directly or indirectly on environmental resources.



Environment and development discourse

Development:

- Development is a process of improving people's economic and social well-being.
- Traditionally, development was strongly associated with **economic growth and increasing income**.
- Modern concepts of development are broader and include:
 - Higher income and employment
 - Better health and education
 - Reduction of poverty
 - Equal opportunities
 - Improved environmental quality
 - Better overall quality of life

Therefore, economic growth is an important component of development, but development is more than economic growth.

Environment and development discourse

Economic Growth and the Environment:

Economic growth refers to an increase in an economy's capacity to produce goods and services over time.

Economic growth can:

- Create employment opportunities
- Increase income
- Encourage investment
- Improve living standards
- Reduce poverty
- However, production requires **energy and natural resources**.
- Increased production and consumption can generate **pollution, waste, and resource depletion**.
- Therefore, uncontrolled economic growth may contribute to environmental degradation.

Environment and development discourse

Sustainability and Sustainable Development:

Sustainability means meeting present needs without compromising the ability of future generations to meet their own needs.

- Sustainable development attempts to balance:
 - Economic development
 - Social well-being
 - Environmental protection
- Natural resources should therefore be used efficiently and responsibly.
- Development should provide benefits today while maintaining environmental resources for future generations.

Environment and development discourse

Why Did Environmental Problems Increase?

Human impacts on the environment increased through major stages of societal development:

Hunter-Gatherer Society → Agricultural Revolution → Industrial Revolution → Information & Globalization Revolution

- **Agricultural Revolution:** increased food production and supported larger populations.
- **Industrial Revolution:** introduced machines, factories, and mass production.
- **Medical innovation:** reduced mortality and increased life expectancy.
- **Technological development:** dramatically increased production and resource consumption.
- **Globalization:** accelerated production, transportation, trade, and consumption.

Environment and development discourse

Our Expanding Ecological Footprint:

- Human societies depend on **natural resources or natural capital**.

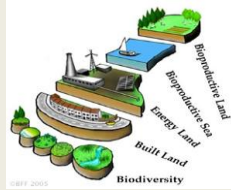
- Population growth increases demand for:

- Food
- Water
- Land
- Housing
- Energy
- Transportation
- Consumer products

- Higher living standards can also increase **per-capita consumption**.

- As ecological footprints increase, natural resources may be depleted faster.

- Greater resource consumption also generates more pollution and waste.



Environmental Issues

Major Environmental Issues

Environmental issues are problems caused or intensified by human activities that negatively affect environmental systems.

Major environmental issues include:

- Air pollution
- Water pollution
- Soil pollution
- Solid waste
- Natural-resource depletion
- Deforestation
- Soil degradation and desertification
- Climate change and global warming
- Natural hazards and environmental degradation

These problems are **interconnected**—one environmental problem can intensify another.

Environmental Issues

Air Pollution:

Air pollution refers to the presence or release of harmful pollutants into the atmosphere.

Major sources:

- Burning fossil fuels
- Industries and factories
- Brick industries
- Motorized vehicles
- Household fuel burning
- Energy production

Major impacts:

- Respiratory problems
- Lung diseases
- Damage to plants and ecosystems
- Contribution to acid rain and climate-related problems



Environmental Issues

Water Pollution:

Water pollution occurs when harmful substances contaminate surface water or groundwater.

Major causes:

- Industrial waste and effluent
- Untreated sewage
- Agricultural fertilizers
- Pesticides and chemicals
- Improper waste disposal

Major consequences:

- Unsafe drinking water
- Water-borne diseases
- Damage to aquatic ecosystems
- Loss of biodiversity
- Contamination of groundwater
- Reduced availability of usable freshwater



Environmental Issues

Soil Pollution and Land Degradation:

Soil pollution can result from:

- Improper solid-waste disposal
- Industrial contaminants
- Excessive fertilizer use
- Agricultural chemicals and pesticides

Soil degradation refers to deterioration in soil quality due to:

- Overuse
- Soil erosion
- Drought
- Poor land management
- Inadequate fertilization

Consequences:

- Reduced soil fertility
- Declining agricultural productivity
- Loss of arable land
- Food-production problems
- Livelihood impacts



Environmental Issues

Solid Waste as an Environmental Problem:

- Increasing consumption generates increasing amounts of **domestic and commercial solid waste**.
- Rapid urbanization makes waste collection and disposal increasingly difficult.
- Poorly managed waste can accumulate in streets, drains, rivers, and open spaces.
- Inadequate waste management contributes to:
 - Disease
 - Water contamination
 - Soil pollution
 - Blocked drainage
 - Poor urban environments
- Landfill sites also have limited capacity.

Environmental Issues

Natural Resource Depletion:

- Human societies depend on natural resources for survival, production, and development.
- Renewable resources:** Water, forests, fisheries, animals, and other biological resources. These can regenerate if used sustainably.
- Non-renewable resources:** Coal, petroleum/oil, and natural gas. These are **finite and cannot be rapidly replaced**.
- Excessive resource use can lead to:**
 - Resource scarcity and reduced future availability
 - Environmental degradation
 - Higher resource costs
 - Resource insecurity
- Even renewable resources such as water can become scarce due to overuse, pollution, and poor management.

Environmental Issues

Deforestation and Ecosystem Loss

- Deforestation** refers to the destruction or conversion of forested land.
- Forests provide essential **ecosystem services**, including:
 - Regulating water supplies and reducing soil erosion
 - Supporting biodiversity and wildlife habitats
 - Providing timber, fuel, food, oils, herbs, and medicines
 - Supporting local livelihoods
- Major causes:** Commercial logging, agricultural expansion, settlement growth, and infrastructure development.
- Major impacts:**
 - Biodiversity and habitat loss
 - Increased soil erosion
 - Disruption of water systems
 - Greater environmental vulnerability



Environmental Issues

Global Warming and the Greenhouse Effect:

- Global warming** refers to the long-term increase in Earth's average temperature, mainly due to an enhanced greenhouse effect.
- How it works:** Solar radiation → Earth's surface warms → Heat is released → Greenhouse gases trap part of the heat
- The natural greenhouse effect keeps Earth warm enough for life; increasing greenhouse gases intensify this warming.
- Major greenhouse gases:** Carbon dioxide (CO₂), Methane (CH₄), and Nitrous oxide (N₂O).
- Major human sources:** Fossil-fuel burning, industries, transportation, agriculture, deforestation, mining, and landfills.

Environmental Issues

Consequences of Global Warming:

Global warming can produce widespread **environmental, social, and economic impacts**, including:

- Temperature:** More frequent heat waves and temperature extremes
- Rainfall:** Changing rainfall patterns and increased drought
- Coastal areas:** Sea-level rise, coastal flooding, and more damaging storms
- Land:** Expansion of dry and degraded areas
- Health:** Worsening air quality and increased health risks
- Agriculture:** Reduced productivity and threats to food security
- Ecosystems:** Habitat disruption and biodiversity loss

Coastal regions are particularly vulnerable because sea-level rise, storms, and flooding can interact and intensify their impacts.

Environmental Issues

Consumerism and Environmental Damage:

Consumption: Use of goods, services, energy, and resources by individuals and societies.

Positive: Higher consumption can improve income, living standards, housing, transport, and services.

Negative: Excessive consumption leads to:

- Greater resource extraction and energy use
- More pollution and waste
- Environmental degradation

Global inequality: Wealthier societies generally consume a greater share of global resources.

Treadmill of Production: Continuous **production → consumption → resource use → pollution and waste**.

Environmental Issues

Environment and Development — Bangladesh Perspective

Bangladesh faces growing environmental pressures from **high population density, rapid urbanization, industrialization, increasing consumption, and climate change**.

Major environmental concerns:

- Coastal inundation and **salinity**
- Changing **temperature and rainfall**
- Drought and seasonal instability
- Deforestation and wetland loss**
- Soil erosion and land degradation
- Degradation of **rivers, lakes, and ponds**
- Industrial pollution and urban waste
- Increasing pressure on **land and water resources**

Thank you all
